DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 5 (AT2)

Visualization Development Exercise

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Question 1 – Chart Selection

1. Comparing total revenue across 6 product categories in one year.
Best Chart Type: Bar Chart (Amounts)
Justification: A bar chart is the best choice for comparing revenue values across different product categories. It clearly shows which category has the highest and lowest revenue.

2. Showing how the salary distribution differs across 4 job levels.
Best Chart Type: Box Plot (Distribution)
Justification: A box plot displays the distribution of salaries, including the median, quartiles, and outliers, making it easy to compare salary variations across the four job levels.

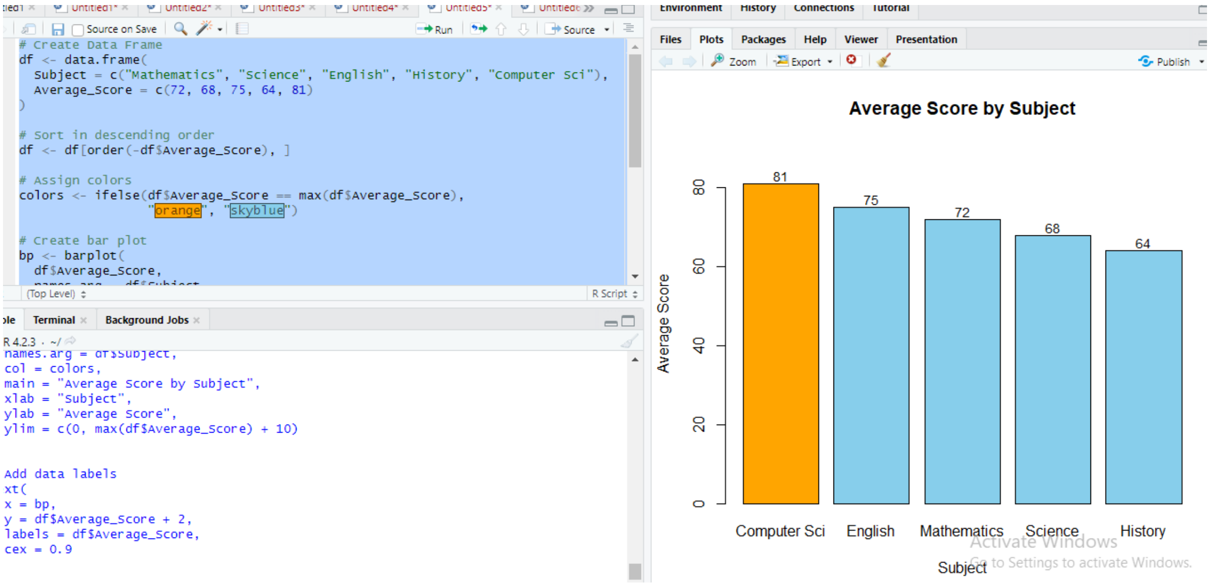
3. Showing market share of 5 competing brands.
Best Chart Type: Pie Chart (Proportion)
Justification: A pie chart is suitable for showing how each brand contributes to the total market share. It highlights the proportion of each brand relative to the whole.

4. Showing the relationship between advertising spend and sales revenue.
Best Chart Type: Scatter Plot (x–y Relationship)
Justification: A scatter plot shows the relationship between two numerical variables and helps identify trends, patterns, or correlations between advertising expenditure and sales revenue.

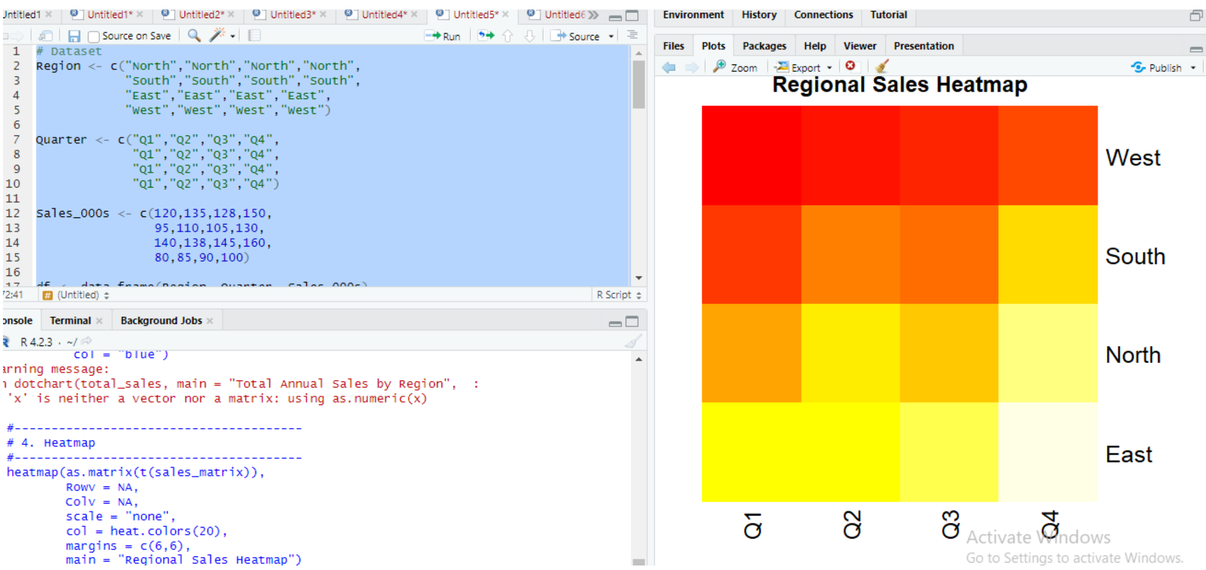
5. Showing monthly website traffic over 3 years.
Best Chart Type: Line Chart (Temporal)
Justification: A line chart is ideal for displaying data over time. It clearly illustrates monthly trends, seasonal patterns, and changes in website traffic over the three-year period.

6. Showing which employees report to which managers in an organization.
Best Chart Type: Network Diagram (Network)
Justification: A network diagram effectively represents reporting relationships between employees and managers. It clearly visualizes the organizational hierarchy and connections between individuals.

QUESTION 2

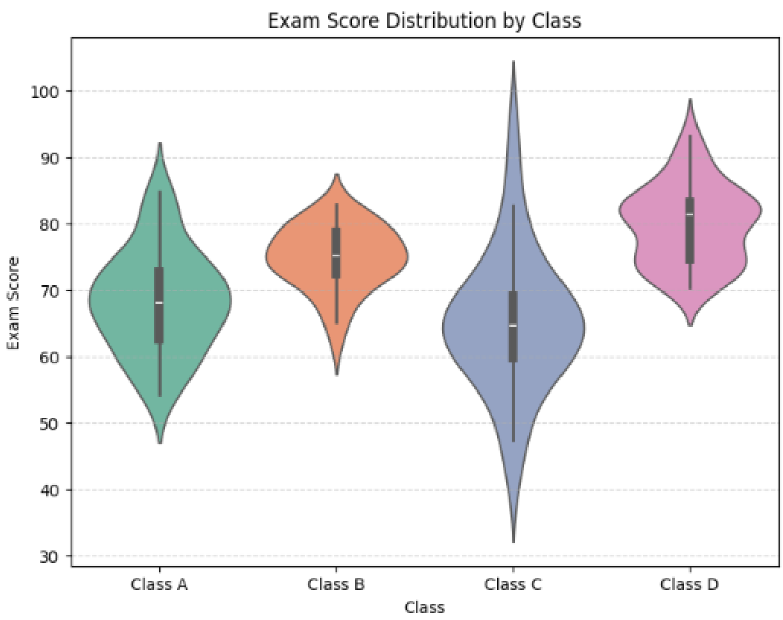


QUESTION 3



QUESTION 4

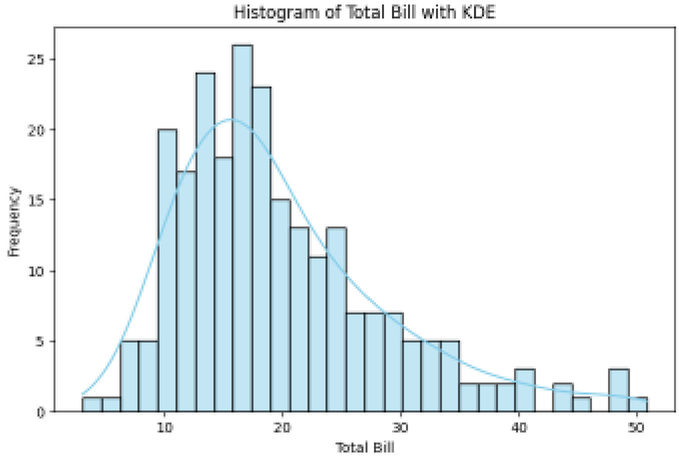
```python
# Install packages (Run once if needed)
# pip install pandas numpy matplotlib seaborn joypy

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns


# ----------------------------
# Generate Synthetic Dataset
# ----------------------------
np.random.seed(42)

class_A = np.random.normal(70, 8, 40)
class_B = np.random.normal(75, 5, 40)
class_C = np.random.normal(65, 12, 40)
class_D = np.random.normal(80, 6, 40)

df = pd.DataFrame({
    'Class': ['Class A']*40 + ['Class B']*40 + ['Class C']*40 + ['Class D']*40,
    'Exam Score': np.concatenate([class_A, class_B, class_C, class_D])
})

# ----------------------------
# 1. Violin Plot with Boxplot
# ----------------------------
plt.figure(figsize=(8,6))

sns.violinplot(
    x='Class',
```



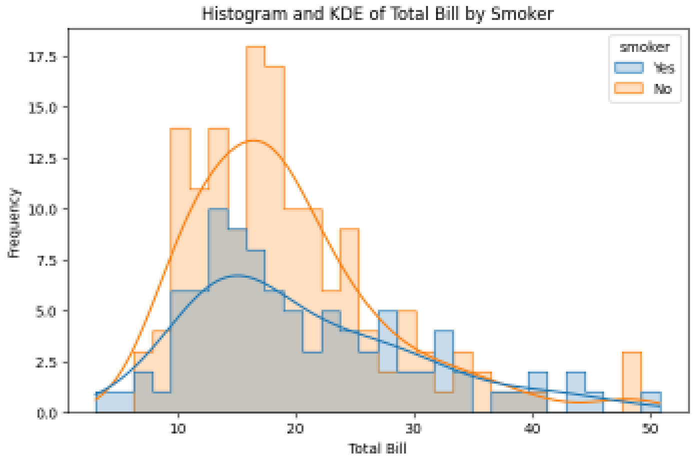
QUESTION 5

```python
# Import libraries
import seaborn as sns
import matplotlib.pyplot as plt

# Load the built-in dataset
tips = sns.load_dataset("tips")

#---------------------------------------
# 1. Histogram with KDE
#---------------------------------------
plt.figure(figsize=(8,5))

sns.histplot(
    data=tips,
    x="total_bill",
    bins=30,
    kde=True,
    color="skyblue"
)

plt.title("Histogram of Total Bill with KDE")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")
plt.show()

#---------------------------------------
# 2. Histogram + KDE by Smoker
#---------------------------------------
plt.figure(figsize=(8,5))

sns.histplot(
    data=tips,
    x="total_bill",
    hue="smoker",
    bins=30,
    kde=True,
```

...



Histogram of Total Bill with KDE
Frequency
Total Bill



Histogram and KDE of Total Bill by Smoker
smoker
Yes
No
Frequency
Total Bill